

VARUN PATHAK

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Technical Skills

Languages	Python, Java, SQL, JavaScript, Bash/Unix, C/C++
Frameworks & Libraries	LangGraph, LangChain, Google ADK, LlamaIndex, OpenAI, Pandas, NumPy, Pydantic, scikit-learn, Matplotlib, asyncio, requests
Platforms & Infrastructure	Google Cloud Platform (GCP), AWS, PostgreSQL, MongoDB, Supabase, ChromaDB, Pinecone, FAISS, pgvector, Jupyter, Apache, Docker, Kubernetes, Git, GitHub, Bitbucket

Experience

Data Science Intern May 2026 – Aug. 2026

The Home Depot Toronto, ON

- Enhanced a production-grade agentic chatbot used by **30,000+ users** leveraging **Google ADK**, **multi-agent orchestration workflows with LLM-backed reasoning**, **MCP (tool-calling)**, and **stateful session management** for internal processes.
- Partnered directly with Directors and cross-functional engineering, product, and operations teams to define requirements, presenting technical insights to VP leadership and **successfully drive 100% of scoped initiatives into production + pilot**.
- Owned 0-to-1 development of scalable LLM applications and reusable agentic microservices across cloud-based GenAI pipelines, integrating emerging AI/ML techniques into production and validating deployment configurations for reliability and performance.
- Created end-to-end **Retrieval-Augmented Generation (RAG)** pipelines with **LlamaIndex**, iterative **prompt engineering**, grounding LLM outputs in domain-specific enterprise data and reducing hallucination risk in production deployments.
- Led **0-to-1** development of a product compatibility POC, **owning a semantic search pipeline** powered by **vector embeddings**, **CloudSQL**, and **Ontology Modeling**; presented findings and secured leadership approval for production.
- Designed and owned **ETL Apache Airflow DAGs** orchestrating ingestion of raw web-sourced data into structured datasets in **GCP BigQuery & PostgreSQL** via **Apache Beam** pipelines, enabling reliable context retrieval for agentic systems.

Instructional Assistant Intern Sept. 2025 – Apr. 2026

McMaster University Hamilton, ON

- Led Python and engineering design (Autodesk Inventor) labs to **600+ students** weekly through labs and tutorials, while providing personalized support during office hours.
- Built a Python automation utility that flagged potential academic dishonesty cases by **scanning 1100+ Autodesk Inventor files** for metadata irregularities, improving grading integrity and **saving manual review time by 80%**.
- Supervised a team of **90+ teaching assistants**, coordinated grading duties, and conducted weekly meetings to maintain instructional consistency and performance.

Machine Learning Engineer Oct. 2024 – Mar. 2026

McMaster Artificial Intelligence Society (MacAI) Hamilton, ON

- Collaborated with a **multidisciplinary research team alongside Dr. Istvan David** to develop and evaluate human-guided multi-agent reinforcement learning (MARL) frameworks, improving coordination and decision-making across agents by **up to 34%** in simulated environments.
- Designed and implemented custom MARL simulation environments in Python to test policy/reward shaping methods, achieving **42% faster convergence compared to unguided baselines**.
- Conducted **over 200 training trials** comparing unguided vs. human-guided MARL, **demonstrating 21% higher collective task success and improved stability** in complex cooperative tasks.

Personal Projects

Stock Forecaster | [🔗](#) – Python, XGBoost, Redis, MLflow Sept. 2026

- Building a probabilistic regression pipeline forecasting next-day/week stock log-returns with calibrated confidence intervals.
- Designed walk-forward validation to prevent lookahead bias, benchmarking **XGBoost** against ARIMA and persistence baselines.

CMU TartanHacks Finalist | [🔗](#) – Python, FastAPI, React Native, Supabase Feb. 2026

- Placed **Top 5** and presented on stage at **Carnegie Mellon University** among **1,100+ participants and 279 projects**.
- Built a passive networking application connecting like-minded people in the background via Bluetooth mesh, using vector-based matching to surface relevant connections in high-density environments.

Education

McMaster University Expected Graduation: May 2028

Software Engineering and Management Co-op (B.Eng.Mgt.) Hamilton, ON

- Dean's Honors List, **CGPA: 3.8/4.0**
- Relevant Courses: Data Structures and Algorithms, Object-Oriented Programming, Databases, Software Design, Agile Methodology, Probability and Statistics, Client Requirements.